

Chapter 1

Introduction

Cryptography is used everywhere today by just about everyone. If you're reading this online no doubt many cryptographic protocols were used to bring the file to your screen. If you're like me and prefer a physical book, even then cryptography was used for the sale, inventory control, and/or bookkeeping of the original vendor. Almost every website today, supports an "https" connection which makes use of public key and private key encryption and authentication protocols. What started as simply an "art" of hiding messages (e.g., encryption), is now a mathematical science, involving rigorous proofs of security, carefully constructed definitions, and the use, if necessary, of minimal, and well-defined, computational assumptions. These are the core pillars of modern day cryptography.

Today, we are generally using classical cryptographic systems, involving the transmission and manipulation of "zeros" and "ones." Messages are classical bit strings and the algorithms used are all running on classical systems, using classical computation. With a few notable exceptions most such classical systems require at least some computational assumptions for their proof of security. Some of the assumptions are fairly "minimal." For instance, the existence of a one-way function (OWF), discussed more later, allows for the secure construction of several private-key cryptographic systems (e.g., encryption, message authentication, pseudo-random functions and generators, just to name some). Other systems, such as public key cryptography, require stronger computational assumptions. While many of the assumptions are believed by many to be true (especially the minimal ones, such as the existence of OWFs), we have no solid rigorous proof and any such proof would require a large breakthrough in computational complexity theory.

While there are some important classical cryptographic protocols that can be proven secure without assumptions, this is usually the exception and often lead to systems that are impractical today. For example, One Time Pad (OTP) encryption can encrypt any message in a way that is "perfectly secret." That is, if you encrypt a message with this system, no adversary, no matter how much time it has to run, and no matter what computational resources it has available (including any possible future technological breakthrough) can learn your message. However, for this system to work, your secret key must be as large, or larger, than the message you're encrypting. Furthermore, the key can only be used once and then it must be discarded. This leads to issues with any practical

implementation: if you want to use OTP encryption, you need to be able to safely distribute, or establish, a very large shared secret key with the party you're communicating with. Today, our secret keys for symmetric key encryption are typically 128 bits or 256-bits long. Such a "small" key can encrypt any size message (e.g., it can be used to encrypt a *1TB* message). On the other hand, if you want to encrypt and send a *1TB* message using OTP, you need to also safely and secretly send or distribute a *1TB* long secret key!

What is this "secret key"? A cryptographic algorithm performs some task, but in general there is something that must be secret. Historically, it was often the case that the *algorithm* that performs the security task was kept secret. However today, we generally make the algorithm public knowledge and instead rely on a small secret number or *key*, which is the secret information required to, for example, send and read messages securely. This notion of making the algorithm public, but relying on a "short" secret token or key is generally known as Kerckhoffs's principle. By following this principle we can design a secure system and use it for multiple applications and parties and then we only need to worry about distributing, and keeping safe, this one random key. Sending this key to a recipient party is called the *key distribution* problem and it is one of the areas that quantum cryptography can greatly help with.

As mentioned above, we have an encryption protocol that is secure without any computational assumptions (i.e., perfectly secret): OTP. However it requires a large secret key. We do *not* have a classical protocol that can distribute a secret key securely without making some form of computational assumption on an adversary's capabilities (i.e., without assuming certain mathematical problems are hard to solve, algorithmically). As it turns out, this is not due to a lack of imagination, but instead it is a mathematical fact that one cannot create a secure secret key distribution protocol that uses only classical information and communication, unless some kind of computational assumption is made on the capabilities of the adversary. In the quantum setting this is not the case, and key distribution is possible without computational assumptions.

Beyond key distribution, there are also other tasks that are either impossible to perform classically, or can only be done by making strong assumptions on an adversary's computational abilities (assumptions which may be invalidated at some later point in time due to some future technological or algorithmic breakthrough). The landscape, however, changes once one augments classical cryptographic systems with quantum information and quantum resources. Suddenly key distribution is possible without the need for computational assumptions. Other tasks such as public key encryption are possible assuming only OWFs (which is impossible in classical cryptography where stronger assumptions are required). Finally, novel cryptographic primitives are suddenly allowed, such as *certified deletion* ? where you can actually have proof that a server deleted your file on request (which is impossible in the classical setting since classical data can always be copied without you knowing).

This book is meant as an introduction to the fascinating field of quantum

cryptography. No prior knowledge of quantum computation or communication is required as these topics will be introduced in sufficient depth in Chapter 2, with additional topics introduced as needed. Of course if the reader is already familiar with quantum computation, Chapter 2 may be skipped (though Chapter 2.6 may be useful as that covers quantum information theory). Of note there is no mention or discussion of quantum algorithms (e.g., Shor’s algorithm); there are, by now, several useful textbooks on such topics. Instead, this book’s focus is on the cryptographic side of quantum science.

A large portion of the book is devoted to so-called *quantum key distribution* (QKD), both protocol design and also security proof methodologies. This is due to the fact that QKD is perhaps the seminal quantum cryptographic application; it is also commercially viable today. Finally, and importantly, many of the methods for analyzing the security of QKD leak into other fields of quantum cryptography, so a solid understanding of QKD (especially their security proofs) is vital to understanding the larger field of quantum cryptography. However, this book will also delve into topics beyond QKD, such as random number generation, bit commitment, alternative security models, and public key encryption, just to list a few topics. This book also primarily focuses on the theoretical study of quantum cryptography; however practical implementation issues are also discussed. It is hoped that the material in this book will serve as a useful foundation to the current student and future researcher, while also serving as inspiration to encourage the further development of this fascinating field.

1.1 QUANTUM COMMUNICATION

Quantum communication involves the sending, and receiving, of *quantum bits* or *qubits*. A classical bit can be in one of two possible states, namely zero “0” or one “1.” A quantum bit can also be in a “zero” or “one” state, which we denote $|0\rangle$ and $|1\rangle$ respectively (that is read “ket zero” and “ket one”). However, a qubit can also be in a *superposition* of all possible states of the form:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (1.1)$$

where $\alpha, \beta \in \mathbb{C}$ such that $|\alpha|^2 + |\beta|^2 = 1$. Thus, unlike a classical bit, there are actually infinitely many states a qubit can be in.

Another difference between qubits and classical bits is in how data is read from them. On receipt of a classical bit, one may read that bit to learn exactly its value; furthermore, reading a classical bit does not disturb it. The fact that you saw the classical bit was in the state “0” does not alter it in anyway. Qubits, however, are different. The act of reading a qubit (called *measuring* a qubit), is a potentially destructive process that produces random output. In particular, given a qubit of the form shown in Equation 1.1, reading this qubit in the so-called *computational basis*, yields an outcome of zero with probability $|\alpha|^2$; otherwise, it yields an outcome of one, with probability $|\beta|^2$ (which, due to the normalization constraint, mentioned earlier, is $1 - |\alpha|^2$).

Not only is the measurement random, it actually causes *state collapse*. That is, if you measure Equation 1.1 and observe “zero”, the state actually becomes $|0\rangle$. The original superposition state of $\alpha|0\rangle + \beta|1\rangle$ is destroyed. If you had actually observed “one” the state collapses to (becomes) $|1\rangle$. Measurements are destructive and irreversible. As it turns out, there are also many ways to measure a qubit (infinitely many, in fact), and one can even make “partial” measurements. It is important to note, though perhaps it is not yet obvious, that a superposition of this form is *not* a “classical mixture.” Namely, the state is not “ $|0\rangle$ ” with probability $|\alpha|^2$ or “ $|1\rangle$ ” otherwise (before measurement). Superposition is something different. Hopefully this will become more clear later in this book.

Note that we said, above, we were measuring in the “computational basis.” As we will see in Chapter 2, qubits are modeled as two-dimensional complex vectors. Thus, one can talk about a basis $\{|v_0\rangle, |v_1\rangle\}$ for this space (we technically require orthonormal bases, but this is not relevant for our high-level discussion, here). The state $|0\rangle$ and $|1\rangle$ are typically called the computational basis, but there are infinitely many other bases. One other important basis is the *Hadamard basis* spanned by states denoted $|+\rangle$ and $|-\rangle$, which are defined, with respect to the computational basis, as:

$$|\pm\rangle = \frac{1}{\sqrt{2}}|0\rangle \pm \frac{1}{\sqrt{2}}|1\rangle. \quad (1.2)$$

In general, if one is measuring in some basis $|v_0\rangle$ and $|v_1\rangle$, the measurement device will return (or observe) one of the two basis states with some probability. State collapse, then, will cause the entire state to collapse to the observed basis state. To compute the probability, one first writes the qubit with respect to that basis (i.e., $|\psi\rangle = x|v_0\rangle + y|v_1\rangle$), and then the probability of observing $|v_i\rangle$ is simply the norm of the scalar coefficient squared. For instance, the probability of observing $|v_0\rangle$ in the above state, would be $|x|^2$; if that is the actual observation, the state collapses to $|v_0\rangle$.

Returning to the computational and Hadamard bases, let’s assume $|\psi\rangle = |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$. Measuring this state in the computational basis will yield an observation of $|0\rangle$ or $|1\rangle$, each with probability 1/2. If one actually observes, say, $|1\rangle$, the state collapses (becomes) $|1\rangle$ and any subsequent measurement will always be $|1\rangle$. On the other hand, if you were to take $|\psi\rangle$ and, instead of measuring in the computational basis, you measure in the Hadamard basis, you will always see $|+\rangle$.

Now, it can be shown, and this will make sense in Chapter 2, that $|0\rangle$ and $|1\rangle$ can actually be written as superpositions (linear combinations, really), of Hadamard states. Namely:

$$|0\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \quad (1.3)$$

$$|1\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle \quad (1.4)$$

Now, let's return to our previous example with $|\psi\rangle = |+\rangle$. Measuring this in the computational basis yields $|0\rangle$ or $|1\rangle$, each with equal probability. Let's say we actually observed $|1\rangle$, and so the state, now, actually is $|1\rangle$. However, since $|1\rangle = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle$, if we now measure this system in the Hadamard basis, the outcome will be either $|+\rangle$ or $|-\rangle$, each with probability $1/2$. Notice that measurements are destructive – if you measure in the “wrong” basis you will get a random result and you cannot “rewind” the measurement to try again. This will be vital, later, when discussing the security of quantum communication. In general, classical data will be hidden in the quantum state by choosing different bases. An adversary, who doesn't know the right basis, will not be able to measure correctly. Any attempt to measure will likely cause state collapse – a disturbance that can be detected.

Additional properties unique to qubits can be proven. For instance, the so-called *no cloning theorem* states that, essentially, one cannot make a copy of an arbitrary, unknown, qubit. Again, this is very different from classical information, where it is very easy to make copies (just hit CTRL-C, CTRL-V for instance!). Given any arbitrary, unknown, quantum state $|\psi\rangle$, you cannot make a perfect copy of it. This may be very annoying as a computer-user, but it is very useful in cryptography: if we send data encoded as random quantum states (i.e., we “hide” our classical data in one or more qubits), an adversary cannot simply make a copy of our data to attack at a later time. Instead, all attacks against quantum cryptographic systems have to be *active attacks*. There is no concept of “copy now, and attack later.” Furthermore, due in part to the destructive nature of measurements, any such attack will leave a fingerprint, or trace, behind, which can be detected.

1.1.1 Entanglement

Besides superposition, *entanglement* is another quantum phenomenon that is highly useful in quantum cryptography. Consider, now, the case where we have two qubits (two physical particles) and they are in the joint state:

$$\frac{1}{\sqrt{2}}|0,0\rangle_{A,B} + \frac{1}{\sqrt{2}}|1,1\rangle_{A,B}. \quad (1.5)$$

Intuitively, from our earlier discussion, if we were to measure *both* particles together, we should see $|0,0\rangle$ with probability $|1/\sqrt{2}|^2 = 1/2$; otherwise we will see $|1,1\rangle$. Furthermore, state collapse causes both particles to collapse to the observed state (either $|0,0\rangle$ or $|1,1\rangle$). However we must keep in mind that these are two separate, physical, systems, and so we can actually separate them (in theory by great distances). Let's send the “A” system to Pluto and keep the “B” system here on Earth. Let's also send someone to Pluto to perform a measurement. If the person on Pluto measures the A system, this is just a single qubit and so can only have a “zero” or “one” outcome. Let's say that this person observed “zero” (which happens half the time). State collapse still occurs;

however, it occurs on the entire joint system, even though the two particles are separated. Thus, if Alice, on Pluto, sees “zero”, her qubit collapses to $|0\rangle$, but also the entire state (including Bob’s particle B here on Earth) also collapses to $|0\rangle$ in this case! Thus, if Bob were to measure his system, he will see exactly the same outcome that Alice sees, no matter how far apart they are.

This is interesting, but if this were the end of the story, entanglement would not be very useful. Indeed, one can simulate the above experiment classically. First, someone, instead of creating two entangled particles as in Equation 1.5, will simply flip a coin, and if its heads, will write the number “0” on two pieces of paper; otherwise the number “1” will be written. Each piece of paper is put in a separate envelop and sealed. One envelope is kept on Earth (the B envelope), while the A envelope is mailed to Pluto (the postage required for such a shipment, would be tremendous, no doubt, but it’s fine to think about these things nonetheless). Now, Alice on Pluto later opens her envelop and she will see a zero or one, that is, as far as she is aware, random. Bob also sees a random zero or one. And, of course, both Alice and Bob “see” the same value, exactly as in the previous quantum experiment. Thus, in this sense, entanglement doesn’t get us anywhere.

However, this is not the entire power of quantum entanglement! In fact, quantum entanglement can produce experimental outcomes that cannot be simulated classically as we did above. As an example, let’s consider the CHSH game ? (named after Clauser, Horne, Shimony, and Holt, and discussed, in more detail, in Chapter 8.1.2). The game consists of two cooperative players, Alice and Bob, and a Referee. The game is such that Alice and Bob will either both win or both lose. First, Alice and Bob are allowed to communicate with one another as much as they like. For instance, they can strategize how to win, they can write down some random information, they can have access to some “randomness in an envelope,” and so on. If Alice and Bob are allowed quantum information, they can also pass along some qubits to one another, including some entangled bits, as in Equation 1.5.

Following this communication stage, Alice and Bob are separated by a distance large enough so that they cannot communicate. That is, Alice and Bob are not allowed to talk any more, but they can of course use any communication they saved from earlier. At this point, the Referee will choose two random bits: x and y ; the value of x will be sent to Alice and the value of y will be sent to Bob. Alice will not know y and, likewise, Bob will not know x . Since they are no longer allowed to communicate, they cannot send their values to each other either.

Given their random values, now, Alice must choose a bit a and Bob must choose a bit b . How they choose these bits can be based on their earlier communication. But, importantly, they are not allowed any additional communication, once they receive their “challenge” x and y . These bits are sent to the Referee.

Alice and Bob both *win* the CHSH game if the following equation is true:

$$a \oplus b = xy, \quad (1.6)$$

where “ $\oplus$ ” is the XOR operation and by xy we mean regular multiplication. Since x and y are both bits, $xy = 1$ only when both x and y are one (otherwise, for all other cases, $xy = 0$). Thus, Alice and Bob win if they choose the same bit (i.e., $a = b$) on all cases where $xy \neq 1$. When $xy = 1$, they must choose different bits to win.

It can be proven that, no matter how much classical communication and randomness they are allowed to share at the beginning of the game (before x and y are chosen), the best chance of winning for Alice and Bob is only 75%. That is, if they play the game multiple times, they can win 3/4 of them on average. However, if Alice and Bob are allowed to share just one entangled pair (Equation 1.5), and this is done before the challenge bits x and y are chosen, suddenly they are able to win with approximately 85% probability. This shows that entanglement is doing something that we cannot simulate with purely classical information and randomness – there is something interesting, and new, here!

We will use entanglement in various quantum cryptographic protocols. QKD is a prime example; however entanglement will also be useful in other cryptographic systems, and also in quantum network settings as entanglement allows for *quantum teleportation*. Quantum teleportation allows an arbitrary qubit state $|\psi\rangle$ to be “teleported” or transferred to another location without actually physically moving it. This is useful in quantum networks, and distributed quantum computation. It can also be used by an adversary, as we discuss later, allowing her to perform rather sophisticated attacks against quantum systems (however, these attacks can still be detected and must still be active!).

1.2 QUANTUM KEY DISTRIBUTION

For a long while, the term quantum cryptography was actually synonymous with quantum key distribution (QKD). Only recently has quantum cryptography developed to such a large extent that QKD is just one of many possible protocols. However, it is still a very important one to understand, and many of the methods and proof techniques used for QKD, extend throughout to other quantum cryptographic protocols.

QKD is a solution to the key distribution problem, discussed earlier. Let’s say we have two parties, Alice and Bob who would like to perform some cryptographic task (e.g., encryption of messages). To do so, they require a *secret key*. This secret key must be random, and must be safely distributed, so that an adversary has no information on it. Today, we use classical systems (such as public key cryptography) to solve this task; however, as discussed earlier, public key systems require computational assumptions on the adversary in order to be proven secure. Should these assumptions fail, the underlying cryptographic primitive is broken. Furthermore, since classical data can be copied, adversaries,

now, are copying data all the time, to attack at a later point. If ever the underlying assumption is found incorrect, any data sent using that protocol would be potentially at risk.

For instance, several classical systems are known to be insecure against large-scale quantum computers. While there are alternative candidate constructions (so-called “post quantum” cryptographic systems), these systems still require computational assumptions (e.g., that it is hard for a computer, either classical or quantum, to solve a particular mathematical puzzle). As discussed, computational assumptions are a requirement for secure key distribution, if one is restricted to classical communication.

QKD solves the key distribution problem, not by relying on computational assumptions, but on physical ones. For example, one requires that certain laws of quantum mechanics are correct, and that user devices are secure and trustworthy. QKD is achieved by encoding classical data onto qubits, observing the error rates in the transmission (which gives a hint as to the attack strategy of an adversary), and then using classical methods to extract a final secret key, even if there is an adversary “snooping” on the data.

To see, at a high level, how this can be done, let’s assume Alice wants to send (or distribute) a secret key $k \in \{0, 1\}^n$ to Bob. We’ll use k_i to mean the i ’th bit of k . To do so, for each $i = 1, \dots, n$, she will choose a random basis, either the computational, or the Hadamard (see Section 1.1), and send k_i encoded in that basis. Essentially, she “hides” the bit information by choosing different qubit encoding strategies.

For example, if her i ’th key bit k_i is a one, and her basis choice for that bit is the Hadamard basis, Alice will send to Bob the qubit state $|-\rangle$ (which is a “one” encoded in the Hadamard basis; a zero in that basis would be $|+\rangle$). Alternatively, if her key bit were zero, and she chose the computational basis, she would send a $|0\rangle$.

The qubit travels to Bob – however, before it arrives, the adversary captures it. Now, due to the no cloning theorem, the adversary, commonly called Eve, cannot simply make a copy to attack later. Instead, she must attack right now. But what can she do? Her goal is to learn k_i and to do so, she must interact with the system in some way. Let’s say she tries to measure – but how should she measure? She can measure in the computational basis or the Hadamard basis (or maybe some other basis, but let’s restrict our attention, for now, on this particular attack strategy). Based on our earlier discussion, if Eve measures in the “wrong” basis (the opposite basis Alice used), then she will get a random outcome. She will also cause a disturbance, and the original state that Alice sent will collapse to a random state in the wrong basis!

Bob, when he finally gets the qubit, will ask Alice for the basis information and Alice can now send this information to Bob. Importantly, this information is not sent in a secret/encrypted manner! Instead, it is sent publicly (one needs authentication, but that detail will be dealt with later in this book). However, thanks to the no cloning theorem, it is “safe” to do so, since Eve already attacked.

Bob, when he learns the correct basis, will perform the correct measurement (which Eve could not do with certainty), and record the outcome. This should match k_i . Of course, if Eve is attacking, Bob's measurement will produce random data (or *noise*). Thus, Alice and Bob will also, occasionally, send “decoys” to check for this kind of attack (i.e., send decoys to check the noise level in the quantum channel, thus determining what an adversary may be up to).

The above is a somewhat high-level description of QKD and there are many subtleties that must be addressed, and will be addressed in this book. However, the main takeaway for now, is, Alice and Bob may “hide” classical data in a qubit state. The qubit cannot be copied to attack later and so an adversary has to attack actively. Any such attack, however, must necessarily create at least some disturbance in the qubit which may be later detected. Importantly, even if an attacker creates disturbance, there are classical methods so as to allow the transmitted information to be distilled into a secret key. If there is too much of a disturbance, then parties simply abort. Thus, in a sense, one either gets a secret key from QKD, or one knows to abort, in which case no sensitive information is lost, because the key has not been used for anything; Alice discards her choices of k_i and chooses again for a later session.

Importantly, just as with quantum computing, quantum cryptography requires both quantum *and* classical information processing. QKD requires quantum operations to distill a “raw key.” But it is classical computing that will distill that raw key into a final secret key. The same is true for any other quantum cryptographic protocol: there is a quantum element, but it must be combined with classical computation, too.

1.3 ALTERNATIVE SECURITY MODELS

QKD does not require computational assumptions for security. Sometimes, however, it is interesting to see what is possible if one were to allow for such assumptions. Indeed, even in the quantum setting, there are several tasks that are impossible to do, securely, without making assumptions. One of the difficulties of working in the field of quantum cryptography is that, first, you must design a suitable protocol and prove it's secure (just as in the classical case), but secondly, you must also show that there is some advantage, whether practical or theoretical, over its classical counterpart. For QKD the advantage is clear. However, when using computational assumptions to build systems, one must be sure that either the application is unique, or that the assumptions required are weaker than what's required classically (otherwise, what's the point of using quantum?).

One fascinating area of recent study is in the development of quantum public key systems. For this, consider a OWF f . Such a function is “easy” to compute, but “hard” to invert. Namely, given x , it is easy to compute $f(x)$. But given y such that $y = f(x)$ for some random x , it is very hard (i.e., computationally inefficient), to find some z such that $f(z) = y$ (z may be the same, or different, from x depending on the function). Whether or not OWFs actually exist is unknown.

However, their existence is necessary for many classical symmetric key cryptographic systems. However, for public key encryption, stronger assumptions are required, at least in the classical setting.

What is a public key encryption protocol? Chapter 3.2 covers this in more detail, but, essentially, such a protocol consists of a secret key sk and public key pk . These two keys are related, somehow, mathematically. The public key is given to everyone (including an adversary), while the secret key is kept secret. Given a copy of the public key, one may encrypt a message. However, only the secret key can be used to decrypt (unlock) the message. This allows many parties to send encrypted messages to one party, and is a vital building block for much of our modern-day communication infrastructure.

That OWFs cannot be used, in a generic fashion, to build a classical public key system has been known for several decades ?? However, rather interestingly, a OWF is sufficient to build a public key encryption protocol, if one also includes some quantum data! In particular, a quantum public key system (discussed more, in Chapter 10), utilizes a classical secret key, and a *quantum* public key. Many copies of the quantum public key are produced by the secret key holder (this does not violate no cloning, since the secret key holder knows the state exactly and should have an efficient quantum algorithm to build a public key state). Anyone with one of the quantum public keys may utilize it to encrypt a message. Only the secret key holder, as in the classical case, can decrypt it.

Computational assumptions are not the only assumptions one may make to an adversary to see what interesting cryptography can be produced. Instead, additional physical assumptions may be made, instead of computational ones. For instance, one may assume that the adversary can do anything within the laws of quantum physics (including having a long run-time to attack), but the adversary has a “noisy” quantum memory. That is, if the adversary stores data as quantum states for too long, it eventually decoheres, and becomes, essentially, random classical bits. Such a physical assumption, which is certainly not too unrealistic today, allows for the development of many additional quantum cryptographic protocols, that are impossible to prove secure in the standard model (even with quantum resources). Oblivious transfer (OT) is one such primitive.

OT is a cryptographic protocol that consists of two parties: Alice and Bob. Unlike key distribution, where Alice and Bob trust one another, here in OT, it is assumed that one of Alice or Bob is the adversary and is trying to cheat the other. In this protocol, Alice has two messages m_0 and m_1 while Bob has a bit b . The goal is for Bob to learn m_b (the message he wants), and no more (i.e., he should not learn m_{1-b}). Furthermore, Alice should never learn Bob’s choice of message b . This task cannot be done, even with quantum bits, without some additional assumptions; if we assume the adversary has “practical” quantum memory, then suddenly it is possible. While it may seem to some that OT is just a “game”, it is actually a vital building block for more advanced and useful, cryptographic protocols (classical and quantum), and so its study is of high importance. We’ll discuss this more, in Chapter 8.3.

The above discussion involved security models where additional assumptions were placed on the adversary (either computational assumptions, or additional physical assumptions). However, one may go in the other direction, and actually give the adversary more capabilities. For instance, the *device independent* model of quantum cryptography assumes that, not only can the adversary do whatever she wants to the transmitted quantum data (within the laws of physics), but she may also have complete control over the source and measurement devices themselves! That is, Alice and Bob do not need to trust their devices - they can be manufactured by an adversary and still you will be able to do some secure cryptography from them.

Take, for instance, random number generation. Now, imagine you buy a box with a button on it, and a single-digit screen. The box is advertised so that, everytime you push the button, a random “zero” or “one” is shown on the screen. You are told, quite confidently, that the box really does what it’s supposed to: namely choose a random bit, totally at random, and totally secret known only to whoever sees the screen. Do you trust it? How can you? An adversarial manufacturer could, for example, pre-program the device with some randomness and the box will simply output the next bit on this “tape.” The manufacturer also has a copy of the pre-programmed randomness, so will know exactly what “random” bits you have. This could be dangerous if these random bits are used for, say, a secret key!

However, if the box is allowed to utilize quantum resources, you can actually test it, yourself, to see that it is indeed giving you true randomness, that no adversary could have pre-programmed or become aware of. Even though you have no idea what’s inside the box, you can perform a statistical test to ensure that the box is somehow using quantum resources the way it’s supposed to. Technically, now, you buy “two” boxes from the adversary and they must be using entanglement between themselves. Essentially, you have these two boxes “play” the CHSH game, described earlier, to ensure that quantum resources, specifically an entangled state of the form shown in Equation 1.5, are being used by the boxes, to produce the random outcomes. Then, due to certain properties of entangled states, one can be sure that the random bits you’re getting are actually safe, even though you never actually look inside the boxes. This will be discussed, more, in Chapter 8.1.2.

1.4 CLOSING COMMENTS

I have tried to make this book accessible to anyone with some basic linear algebra and probability theory. The topics in quantum mechanics needed to at least understand the fundamentals are all covered, as are also those topics needed in classical cryptography. Having an appreciation, and understanding, of both is, I think, required to fully understand quantum cryptography. The field, however, is quite vast at this point, and so I had to be selective in which topics I covered. Not everything will be here. However, the knowledge gained should provide the

reader with enough material to be able to understand, at least at a high level, other topics in quantum cryptography that were not covered here.

There is sometimes a divide in some corners of the world, between those who study “quantum” cryptography (such as QKD), and those who study “post-quantum” cryptography (which is classical cryptography, with stronger mathematical assumptions). It is my belief that both fields have strong merits, and I will not try to unduly promote one system as more advantageous than another. QKD, and quantum cryptography in general, has several advantages over classical cryptography, including post-quantum cryptography; it also has several disadvantages, which will be discussed throughout this book. The same is true for post-quantum systems. In reality, it will likely be that both systems should be utilized side-by-side and such research is being investigated by several groups. This book will only cover quantum cryptography and will not touch on any topics in post-quantum. The methods and protocols are very different, that books in both fields are required, just as both fields are required to secure our future communication infrastructure.

It is my hope that this book will serve as an inspiration for some to become more interested in quantum cryptographic research. I started my own career in quantum cryptography purely by accident: as a graduate student, I would wander the library often, looking for something interesting to read. One day, a book with an interesting title caught my eye, “Automatic Quantum Computer Programming: A Genetic Programming Approach” by Lee Spector. I took out the book, and read it over my summer break and was immediately hooked. I hope this book will also hook a few into further studying and developing this field.